

Project Summary Report

Date: 25/08/2026

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Dataset: HealthConnect Appointment Dataset

1. Business Problem

As a Data analytics intern, I am addressing two interconnected business problems at HealthConnect Clinic being appointment no-show and difficulty understanding no-show. While my main report documents the full context of both problems, my specific focus through analytics is to identify patterns and factors associated with appointment no-shows, enabling the clinic to understand why patients miss appointments and take action to reduce missed appointments.

2. Resources Used

HealthConnect_Appointment_Data.csv: primary dataset with 5,000 appointment records

HealthConnect_Data_Dictionary.xlsx: Variable definitions & data types

Python: missing value identification and profiling

Power BI: initial column quality assessment & profiling

3. Key Observations

The dataset has 5,000 appointment records from 2024 – 2026 with 18 variables. The dataset had 150 missing values in total from distance_to_clinic_kn with 90 missing values and waiting_time_minutes with 60 missing values. Date column also needs to be converted to dates. Our target category (appointment_outcome) has 3 categories being attended, no-show and cancelled.

4. Proposed Approach

Data preparation: Handle missing values, convert dates, validate calculated fields

Exploratory Analysis: Create visualizations, identify no-show patterns

Hypothesis Testing: Statistically test relationships between variables and no-show outcomes

Dashboard development: Build KPI cards, design layout, create interactive visuals

Reporting: summarize findings, provide recommendations

5. Key Considerations

The dataset is fictional and hence might not provide real insights

Missing values require careful handling to avoid data skewing

No data regarding rescheduling and this data may impact understanding of cancellation patterns

6. Documentation for next phase

The next phase of my analysis will focus on.

Complete Data Preparation

- Replace all missing values and validate fields.
- Document all data transformations.

Detailed Exploratory Data Analysis

- Create summary statistics for all variables.
- Build visualizations exploring each business question.
- Identify patterns and trends in no-show behavior.

Statistical Analysis

- Outline hypotheses.
- Identify which variables will be tested.

Dashboard development

- Define Dax measures needed.
- Design the executive summary.

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